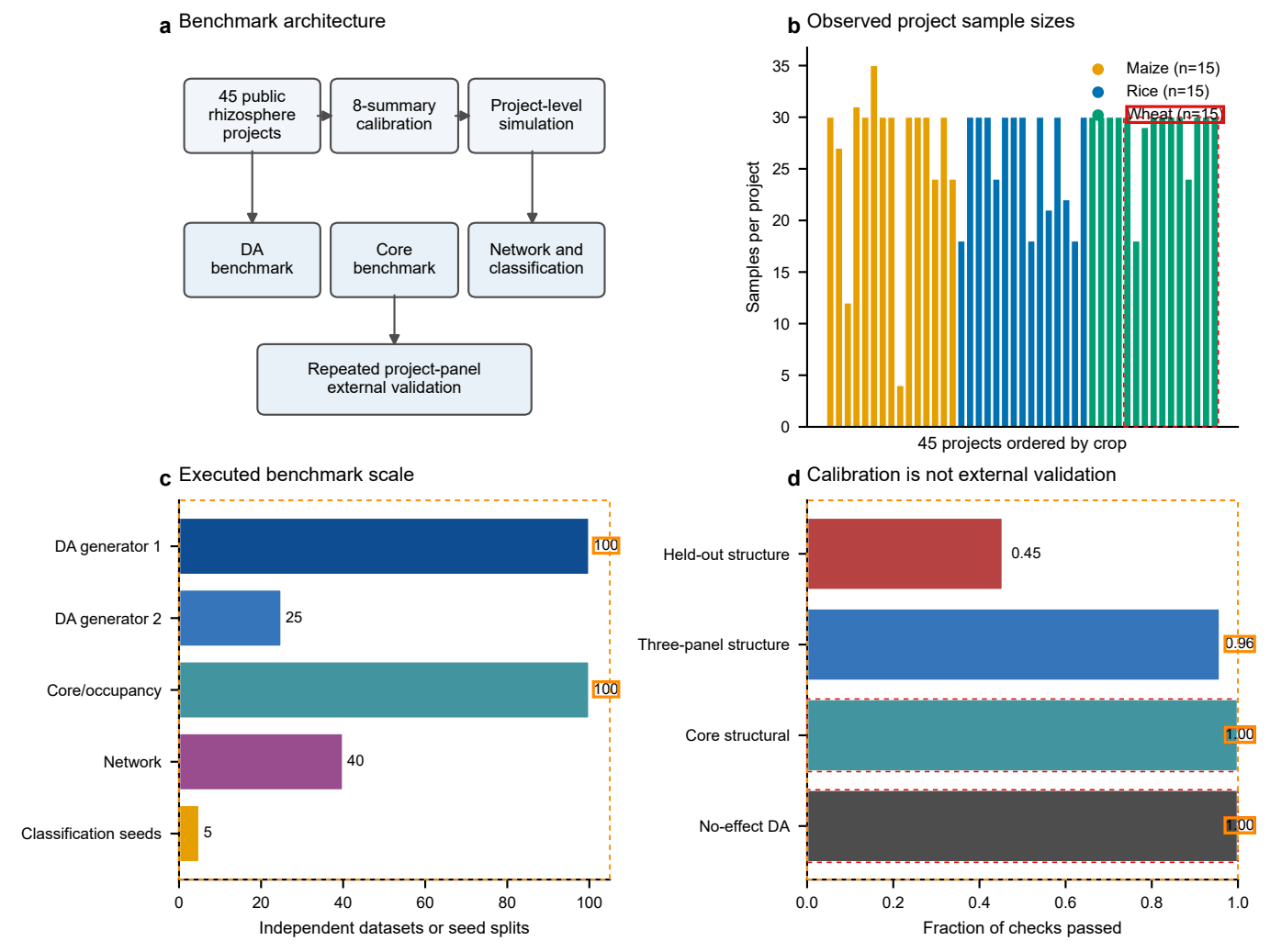
a Benchmark architecture
45 public rhizosphere projects
8-summary calibration
Project-level simulation
DA benchmark
Core benchmark
Network and classification
Repeated project-panel external validation
b Observed project sample sizes
Maize (n=15)
Rice (n=15)
Wheat (n=15)
Samples per project
45 projects ordered by crop
c Executed benchmark scale
DA generator 1
DA generator 2
Core/occupancy
Network
Classification seeds
100
25
100
40
5
Independent datasets or seed splits
d Calibration is not external validation
Held-out structure
Three-panel structure
Core structural
No-effect DA
0.45
0.96
1.00
1.00
Fraction of checks passed